(ii) Explaining steady state sol-n:

We have $\boxed{u=0} \Rightarrow \boxed{\nabla u = \underline{0}}$ & of course $\nabla^2 u = 0$,

for again $\underline{x} \in \mathbb{R}^3$,

$$(9) \quad \begin{cases} u_{xx} + u_{yy} + u_{zz} = 0 \\ u_x = u_y = u_z = 0 \\ u(x,y,z) = 0 \end{cases}$$

which is a steady-state $\Rightarrow$ objects' temperature is always zero ($u=0$) & it is not changing over time ($\nabla u = 0$).

$\blacksquare$

Prob. 3. (ZaT, Ch 6, 1.3 (mod-d)).

Let u be in $\underbrace{C^2(\Omega) \cap C^1(\bar{\Omega})}$, where Ω is a regular domain in $\mathbb{R}^n$. Suppose that

?2 why needed?

$$(1) \quad \nabla^2 u = 0 \qquad \Omega$$

$$(2) \quad \frac{\partial u}{\partial \underline{n}} = 0 \qquad \partial\Omega$$

WTS that (i), is a const. in $\bar{\Omega}$. The text says Ω is a "normal bounded domain" - this is the same as being "regular". Assume $\underline{x} \in \mathbb{R}^3$ & notes are the same with previous Prob. 2, (ii) Explain steady-state.

Sol-n. [Preamble].

Let $\underline{x} \in \mathbb{R}^3$ & the following calculations hold for $\underline{x} \in \mathbb{R}^n$, WLOG.

Using the Green's 1st identity with $w = u$ like in previous problem 2.

$$\int_\Omega u \cdot \underline{\nabla}^2 w \, d\underline{x} = \int_{\partial\Omega} u \cdot \frac{\partial w}{\partial \underline{n}} \, ds - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} w \, d\underline{x} \Rightarrow$$

$$\Rightarrow \int_\Omega u \cdot \underbrace{\underline{\nabla}^2 u}_{=0} \, d\underline{x} = \int_S u \cdot \underbrace{\frac{\partial u}{\partial \underline{n}}}_{=0} \, ds - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} u \, d\underline{x} \Rightarrow \text{So}$$

$$\int_\Omega 0 \, d\underline{x} = \int_S 0 \, ds - \int_\Omega (\underline{\nabla} u)^2 \, d\underline{x} \Rightarrow$$

(3) $$\int_\Omega (\underline{\nabla} u)^2 \, d\underline{x} = 0$$ [By the hint given in Prob. 2 which is about the integral of positive fn] $\Rightarrow$

$$\Rightarrow (\underline{\nabla} u)^2 = 0 \Rightarrow \langle u_x, u_y, u_z \rangle \cdot \langle u_x, u_y, u_z \rangle = 0$$

$$\Rightarrow u_x^2 + u_y^2 + u_z^2 = 0 \Rightarrow$$

(4) $u_x = 0$ $\Downarrow$ (5) $u_y = 0$ (6) $u_z = 0$

(7) $\boxed{u(x,y,z) = C_1(y,z).}$ If we substitute this (7) into (5) & (6) above respectively, we get

$$u_y = \frac{\partial}{\partial y}[C_1(y,z)] = 0 \Rightarrow u = C_1(y,z) = C_2(z) \quad \text{otherwise}$$

we never get $C_{1y} = 0$ unless $C_1(y,z) = C_2(z)$. Similarly Plugging this $u = C_2(z)$ into (6) gives

$$u_z = \frac{\partial}{\partial z}[C_1(z)] = 0 \Rightarrow C_1(z) \equiv C_3 = \text{const.}$$

Again if $u = C_1(z) \neq C_3$, we never get $u_z = C_{1z}(z) = 0$.

(8) Hence $\boxed{u(x,y,z) \equiv C_3}$ in Ω.

So by (8) & (2) we have

(8) $u(x,y,z) = C_3$ in Ω

(2) $\frac{\partial u}{\partial \underline{n}} = \underline{\nabla} u \cdot \underline{n} = 0$ in $\partial\Omega$

Of course $\underline{n} \neq \underline{0}$ (outward normal $\neq \underline{0}$), then we must have 2 cases for the <u>dot prod.</u> in (2) $\equiv 0$:

Case (i) Either $\underline{\nabla} u \perp \underline{n}$ always on $\partial\Omega$, (i.e. $|\underline{\nabla} u| \cdot |\underline{n}| \cdot \cos(\frac{\pi}{2}) \equiv 0$)

$\Rightarrow |\underline{\nabla} u| \cdot \underbrace{|\underline{n}|}_{\neq 0} \cdot \cos(\frac{\pi}{2}) = 0 \Rightarrow$ leads to the possibility

(9) $|\underline{\nabla} u| \neq 0$ on $\partial\Omega$. $\Rightarrow$

$\sqrt{\underline{\nabla} u \cdot \underline{\nabla} u} \neq 0 \Rightarrow$

$\underline{\nabla} u \cdot \underline{\nabla} u \neq 0 \Rightarrow$

(10) $\boxed{\underline{\nabla} u \neq \underline{0}}$ on $\partial\Omega$. However, then

$$\begin{cases} (8) & u(x,y,z) = C_3 \text{ in } \Omega \\ (10) & \underline{\nabla} u = \langle u_x, u_y, u_z \rangle \neq \underline{0} \text{ in } \partial\Omega \end{cases} \Rightarrow$$

$$(11) \begin{cases} u(x,y,z) = C_3 \text{ in } \Omega \\ u(x,y,z) \equiv f(x,y,z) \neq \text{const. on } \partial\Omega \end{cases} \left(\text{for } \underline{\nabla} u = \langle f_x, f_y, f_z \rangle \neq \underline{0}\right)$$

But then the system (11), by which u is given piecewise, violates the given behaviour of u:

note the closure!

$$u \in C^2(\Omega) \cap C^1(\bar{\Omega}). \Rightarrow \boxed{u \in C^1(\bar{\Omega})}.$$

So by (11) $u = C_3$ in Ω, $u = f(x,y,z) \neq C_3$ on $\partial\Omega$, $\Rightarrow$

(12) $\boxed{u \notin C^1(\bar{\Omega})}$, (i.e. u is not continuous on $\bar{\Omega}$).

So with case (i) $\Rightarrow$ system (11) $\Rightarrow$ $u \notin C^1(\bar{\Omega})$ $\lightning$ this case never happens.

Case (ii) or $\underline{\nabla} u = \underline{0}$ always on $\partial\Omega$ $\left(\text{for } \frac{\partial u}{\partial \underline{n}} = \underline{\nabla} u \cdot \underline{n} = 0 \text{ to work!}\right)$.

Then,

(13) $\boxed{\underline{\nabla} u = \underline{0}} \Rightarrow \langle u_x, u_y, u_z \rangle = \underline{0}$ on $\partial\Omega$,

which implies $u(x,y,z) \equiv D = \text{const.}$ on $\partial\Omega$. So

(8) $u(x,y,z) = C_3$ in Ω

(13) $u(x,y,z) = D$ on $\partial\Omega$ $\left(\text{as } \underline{\nabla} u = \langle \frac{\partial}{\partial x}(D), \frac{\partial}{\partial y}(D), \frac{\partial}{\partial z}(D) \rangle = \underline{0}\right)$ works.

Of course this (8) + (13) system matches the given behaviour of u:

(14) $\boxed{u \in C^2(\Omega) \cap C^1(\bar{\Omega})}$

as long as $D \equiv C_3$, otherwise $\begin{cases} u = C_3 & \text{in } \Omega \\ u = D & \text{in } \partial\Omega \end{cases} \Rightarrow$

u has a jump close to the boundary $\Rightarrow$ u is not continuous on $\bar{\Omega}$. Hence

(15) $\begin{cases} u(x,y,z) = C_3 & \text{in } \Omega \\ u(x,y,z) = C_3 & \text{on } \partial\Omega \end{cases} \Rightarrow u \in C^2(\Omega) \cap C^1(\bar{\Omega})$

establishing the constant behaviour of u on $\bar{\Omega}$. $\blacksquare$

(ii) Explanation of the solution for the given problem:

$\begin{cases} u = C_3 & \text{in } \Omega \\ \frac{\partial u}{\partial \underline{n}} = 0 & \text{on } \partial\Omega \end{cases} \Rightarrow$ the boundary is insulated, then with time increase in the domain, the temperature stays <u>const</u> $= C_3$. $\blacksquare$